

Code-Layout, Readability and Reusability

Date September 2026
Team ID xxxxxx
Project Name EduMind AI - Smart AI-Powered Learning Assistant
Maximum Marks 3 Marks

Step 1: Project Code Structure

EduMind AI is organized into separate files and modules so that the backend, AI integration and learning features can be maintained independently.

File / Module	Purpose	Readability / Reusability Practice
main.py	FastAPI application, routes and application startup.	Clear route functions and descriptive endpoint names.
gemini_client.py	Gemini API client creation and AI content generation.	Centralized AI integration reused by feature modules.
qna.py	Question-answering prompt and response generation.	Single-purpose function with clear input/output.
explanation_module.py	Beginner-friendly concept explanations.	Feature isolated in its own reusable module.
quiz_module.py	Quiz generation workflow.	Reusable generation function for different topics.
notes_module.py	Study-notes generation.	Consistent prompt structure and clear formatting.
summary_module.py	Content summarization.	Independent summary function for reuse.
learning_path.py	Learning-path recommendations.	Topic and level passed as reusable parameters.
templates / static	User interface files and static assets.	Presentation layer kept separate from backend logic.

Step 2: Readability Guidelines

The code uses descriptive names, comments where useful, modular functions, consistent formatting and simple control flow. Each feature has a clear responsibility, making the project easier for team members to understand.

Step 3: Reusability Guidelines

Common Gemini API communication is centralized in `gemini_client.py`. Individual feature modules call the same generation function instead of duplicating API setup code. This reduces duplication and makes future model or configuration changes easier to manage.